

Departement Elektriese, Elektroniese en  
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and  
Computer Engineering

## Semestertoets 1

Kopiereg voorbehou

## Semester Test 1

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**Module EBN122**  
**Elektrisiteit en Elektronika**  
25 Augustus 2016

**Module EBN122**  
**Electricity and Electronics**  
25 August 2016



UNIVERSITEIT VAN PRETORIA  
UNIVERSITY OF PRETORIA  
YUNIBESITHI YA PRETORIA

### Eksamensinligting:

#### Exam information:

|                                                                                                        |                         |                                                  |               |
|--------------------------------------------------------------------------------------------------------|-------------------------|--------------------------------------------------|---------------|
| Maksimum punte:<br><i>Maximum marks:</i>                                                               | 46                      | Volpunte:<br><i>Full marks:</i>                  | 45            |
| Duur van vraestel:<br><i>Duration of paper:</i>                                                        | 90 minute<br>90 minutes | Oopboek / toeboek:<br><i>Open / closed book:</i> | Toe<br>Closed |
| Totale aantal bladsye (hierdie blad ingesluit):<br><i>Total number of pages (including this page):</i> | 15 + Answer sheet       |                                                  |               |

### BELANGRIK- IMPORTANT

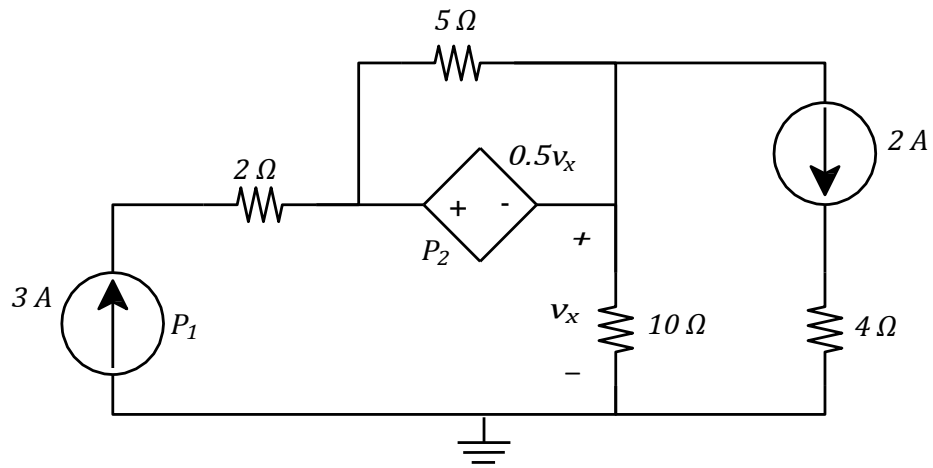
- Die eksamenregulasies van die Universiteit van Pretoria geld.  
*The examination regulations of the University of Pretoria apply.*
- Alle vrae moet op die ANTWOORDBLAD beantwoord word.  
*All questions must be answered on the ANSWER SHEET.*
- Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.  
*Students may do their own rough calculations on the question paper – the question paper will not be taken in.*
- Antwoorde moet eenhede insluit!  
*Answers must include units!*
- Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**  
***Final answers must be written as real numbers, and not as fractions.***

|                                   |                 |
|-----------------------------------|-----------------|
| Dosent(e):<br><i>Lecturer(s):</i> | 1. D le Roux    |
|                                   | 2. J. Joubert   |
|                                   | 3. R dos Santos |

## **VRAAG/QUESTION 1 [8]**

|                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Question 1.1 [2]</b><br>The kitchen light in a household is used for 8 hours every day per year (365 days). If the lightbulb was a 100 W candescent lightbulb, what is the savings over a one year period if the lightbulb is replaced with a 20 W LED light? Assume electricity costs R1.50 per kWh. | <b>Vraag 1.1 [2]</b><br>‘n Gloeilamp in ‘n kombuis word vir 8 ure elke dag van die jaar (365 dae) gebruik. Indien die gloeilamp met ‘n gradering van 100 W vervang word met ‘n LED lamp van 20 W, wat is die besparing oor ‘n een jaar periode? Aanvaar elektrisiteit kos R1.50 per kWh. |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Question 1.2 [2]</b><br>A 5V battery is connected in series with an element. The current exiting the positive terminal of the battery decreases linearly from 20 mA to 5 mA between $t = 0$ s and $t = 20$ s, while the voltage remains constant. What is the energy associated with the battery? | <b>Vraag 1.2 [2]</b><br>‘n 5 V battery word in serie met ‘n element gekoppel. Die stroom wat by die positiewe terminal van die battery uitvloei val liniêr van 20mA tot 5 mA tussen $t = 0$ s en $t = 20$ s. Wat is die energie geassosieer met die battery? |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Question 1.3 [4]**Determine  $P_1$  and  $P_2$  if  $V_X = 10\text{ V}$ .**Vraag 1.3 [4]**Bepaal  $P_1$  en  $P_2$  indien  $V_X = 10\text{ V}$ .

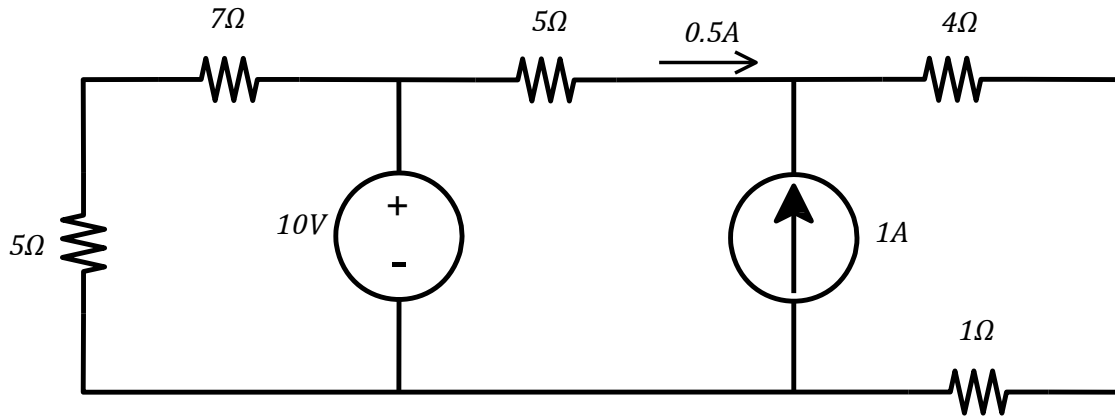
## VRAAG/QUESTION 2 [16]

### Question 2.1 [4]

- a) Determine the power associated with the current source.
- b) Determine the power associated with the  $7\Omega$  resistor.

### Vraag 2.1 [4]

- a) Bepaal die drywing ge-assosieer met die stroombron.
- b) Bepaal die drywing ge-assosieer met die  $7\Omega$  weerstand.

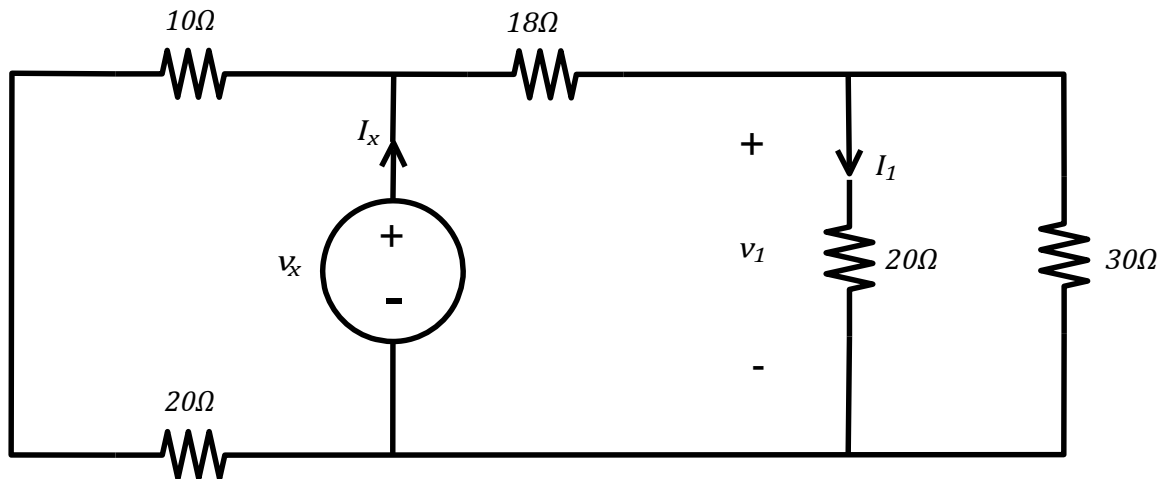


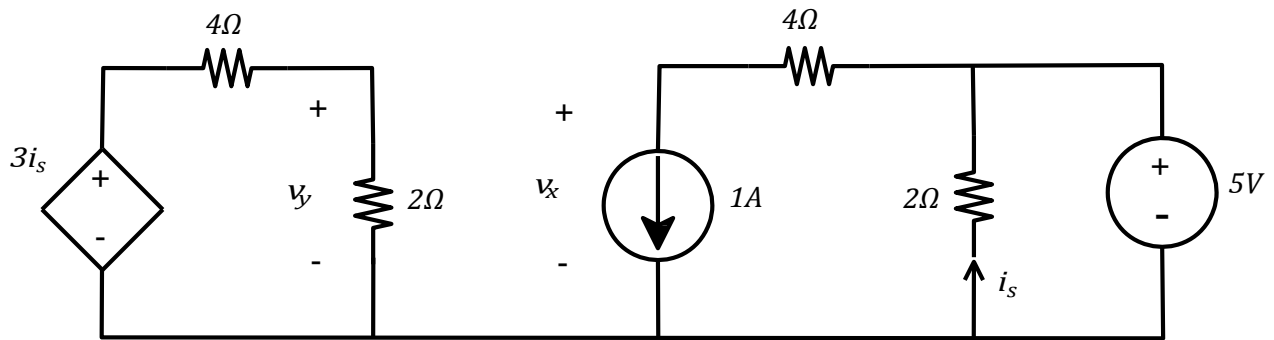
**Question 2.2 [4]**

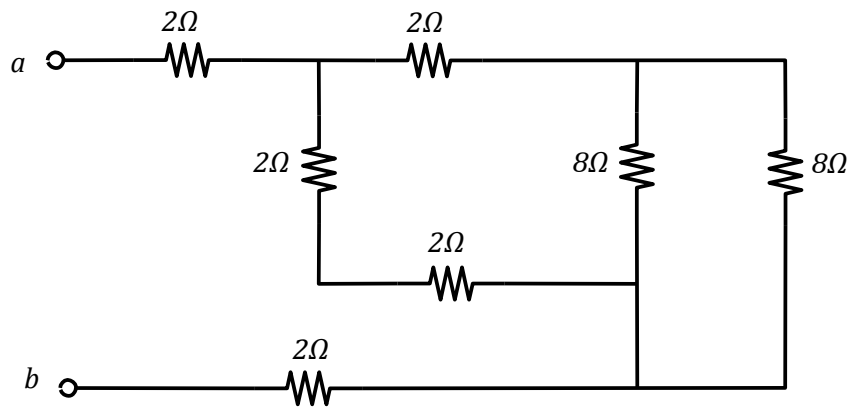
- a) Determine  $v_1$  if  $v_x = 15\text{V}$ .  
b) Determine  $I_1$  if  $I_x = 3\text{A}$ .

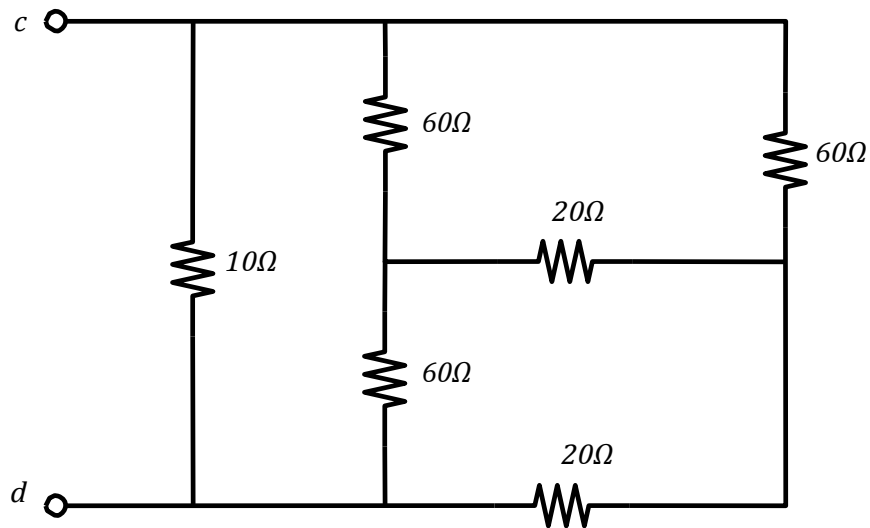
**Vraag 2.2 [4]**

- a) Bepaal  $v_1$  as  $v_x = 15\text{V}$ .  
b) Bepaal  $I_1$  as  $I_x = 3\text{A}$ .



**Question 2.3 [4]**Determine  $V_x$  and  $V_y$ .**Vraag 2.3 [4]**Bepaal  $V_x$  en  $V_y$ .

**Question 2.4 [2]**Find the equivalent resistance wrt terminals  $a$ - $b$ .**Vraag 2.4 [2]**Bepaal die ekwivalente weerstand tov terminale  $a$ - $b$ .

**Question 2.5 [2]**Find the equivalent resistance wrt terminals  $c$ - $d$ .**Vraag 2.5 [2]**Bepaal die ekwivalente weerstand tov terminale  $c$ - $d$ .



### QUESTION/VRAAG 3 [22]

**Question 3.1 [4]**

Use nodal analysis to set up two equations of the following form:

$$2V_1 + aV_2 = b$$

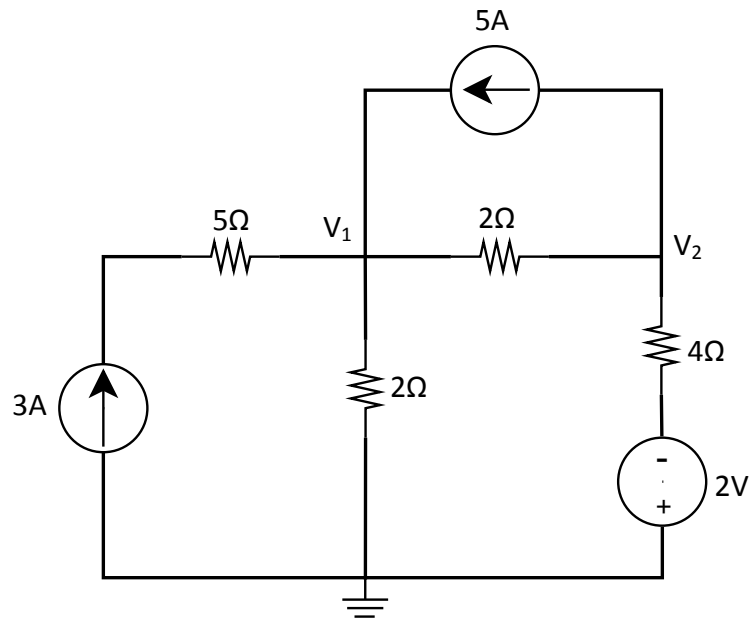
$$2V_1 + cV_2 = d$$

**Vraag 3.1 [4]**

Gebruik lusanalyse om twee vergelykings van die volgende vorm op te stel:

$$2V_1 + aV_2 = b$$

$$2V_1 + cV_2 = d$$



**Question 3.2 [4]**

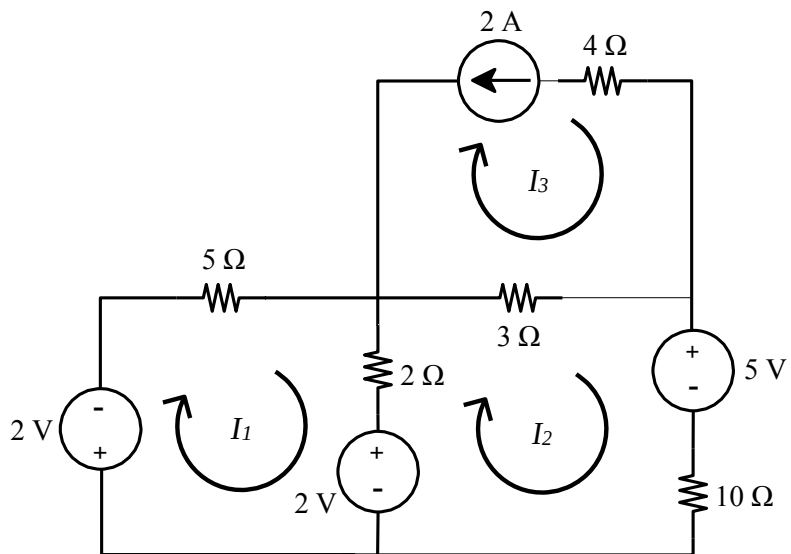
Use mesh analysis to set up two equations of the following form:

$$\begin{aligned} eI_1 - 2I_2 &= f \\ -2I_1 + gI_2 &= h \end{aligned}$$

**Vraag 3.2 [4]**

Gebruik lusanalyse om twee vergelykings van die volgende vorm op te stel:

$$\begin{aligned} eI_1 - 2I_2 &= f \\ -2I_1 + gI_2 &= h \end{aligned}$$



**Question 3.3 [4]**

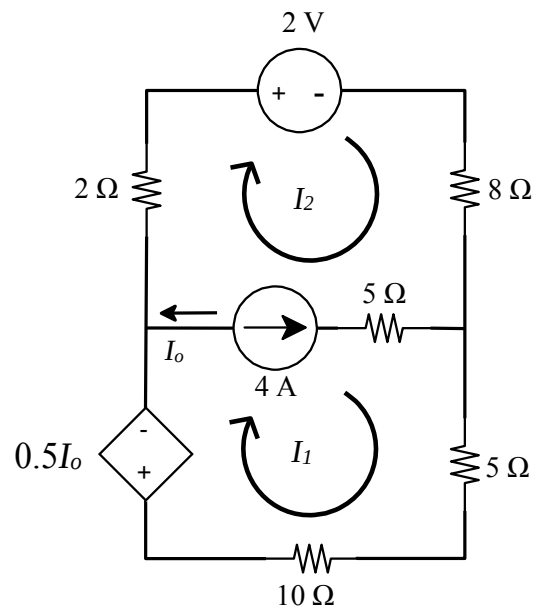
Use mesh analysis to set up two equations of the following form:

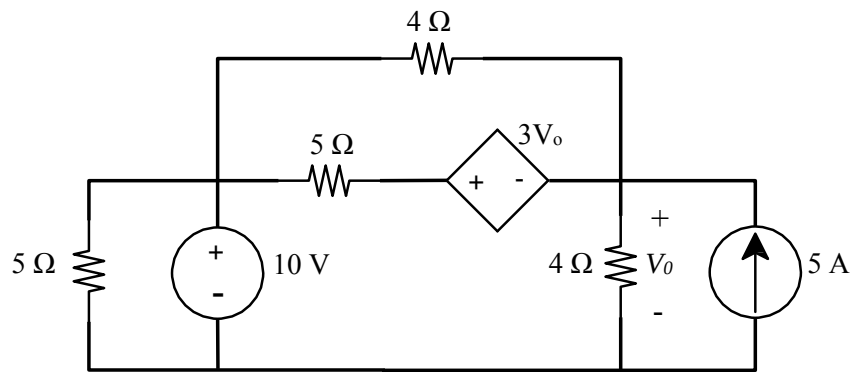
$$\begin{aligned} I_1 + mI_2 &= n \\ pI_1 + qI_2 &= -1 \end{aligned}$$

**Question 3.3 [4]**

Gebruik lusanalyse om twee vergelykings van die volgende vorm op te stel:

$$\begin{aligned} I_1 + mI_2 &= n \\ pI_1 + qI_2 &= -1 \end{aligned}$$



**Question 3.4 [2]**Solve for  $V_o$ .**Vraag 3.4 [2]**Los op vir  $V_o$ .

**Question 3.5 [2]**

Determine  $\alpha$ , if  $v_2 = -5$  V and  $v_3 = 0.5$  V.

$$\begin{bmatrix} 2 & 0 & 5 \\ 0 & 1 & 2\alpha \\ 2 & 0 & 3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ 3 \end{bmatrix}$$

**Question 3.5 [2]**

Bepaal  $\alpha$ , as  $v_2 = -5$  V en  $v_3 = 0.5$  V.

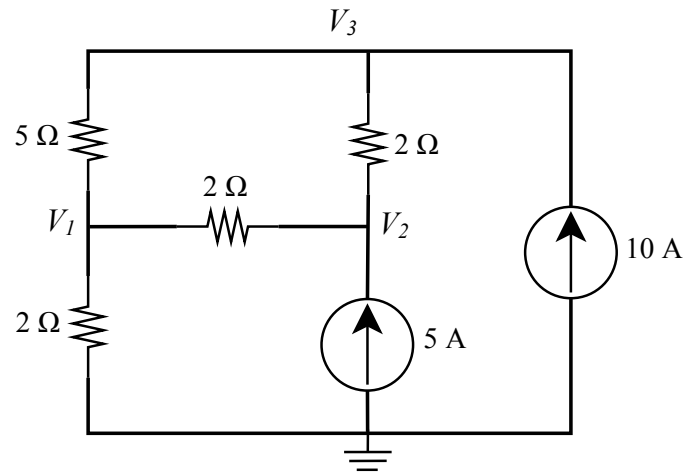
$$\begin{bmatrix} 2 & 0 & 5 \\ 0 & 1 & 2\alpha \\ 2 & 0 & 3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ 3 \end{bmatrix}$$

**Question 3.6 [3]**

Use inspection to set up a matrix equation in terms of the node voltages.

**Vraag 3.6 [3]**

Stel 'n matriksvergelyking dmv inspeksie op in terme van die node-spannings.



$$\begin{bmatrix} & & -0.2 \\ & 1 & \\ -0.2 & & \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} \\ \\ \end{bmatrix}$$

**Question 3.7 [4]**

Given:  $\beta = 100$ ,  $I_E = 3\text{ A}$  and  $V_{BE} = 0.7\text{ V}$ .

- a) Determine  $V_S$ .
- b) Determine  $V_{CE}$ .

**Vraag 3.7 [4]**

Gegee:  $\beta = 100$ ,  $I_E = 3\text{ A}$  en  $V_{BE} = 0.7\text{ V}$

- a) Bepaal  $V_S$ .
- b) Bepaal  $V_{CE}$ .

